

Fubin Dou

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Education

University of Science and Technology of China, School of Mathematical Sciences
B.S. in Computational Mathematics
Anhui, China
Sept. 2023 – Jun. 2027 (Expected)

Honors and Awards

- Outstanding Student Scholarship, 2025

Research Interests

My research interests lie at the intersection of applied mathematics, artificial intelligence, computer vision, and computer graphics, with a focus on dynamic 3D reconstruction, neural rendering, physics-constrained deformable modeling, optimization and inverse problems, scientific computing, and AI for Science.

Publications

- Haoqin Hong*, Ding Fan*, **Fubin Dou**, Zhi-Li Zhou, Haoran Sun, Congcong Zhu, Jingrun Chen. *Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field*. Association for the Advancement of Artificial Intelligence (AAAI), 2026. **Accepted**.

Research Experience

SCAI Lab, University of Science and Technology of China
Anhui, China Nov. 2024 – Present
Advisor: Prof. Jingrun Chen, Department of Mathematics, USTC
Participated in research on a physics-informed deformable 3D Gaussian Splatting framework that introduces continuum mechanics into dynamic scene reconstruction. I worked on the static–dynamic decoupling and dynamic mask modules, separating moving foreground regions from static backgrounds to alleviate camera/object motion ambiguity and improve both static background geometry and dynamic foreground motion consistency.

Academic Projects

Differentiable Physics-Driven Spring-Mass System and Inverse Reconstruction of Dynamic Scenes University of Science and Technology of China
Project on deformable dynamic scene modeling and inverse reconstruction Built a differentiable physics simulation framework based on spring-mass systems and embedded the simulation process into an optimization loop to infer scene states and material parameters from observed dynamic sequences. The project connects discrete spring systems with continuum mechanics, fits Neo-Hookean elastic energy to a discrete topology through energy matching, estimates physical parameters such as Young’s modulus and Poisson’s ratio, and implements simulation and automatic differentiation workflows with Taichi. It further uses differentiable rendering and physics losses to optimize dynamic deformation, providing an optimization-based approach for recovering geometry, motion, and physical properties of deformable objects from visual observations.

Relevant Coursework

Real Analysis (4.0)	Complex Analysis (4.0)	Functional Analysis (4.0)
Multivariable Calculus (4.0)	Mathematical Statistics (4.0)	Numerical Algebra (4.0)

Skills

- Programming Languages:** C, C++, Python, $\text{\LaTeX}$
- Deep Learning and Scientific Computing:** PyTorch, Taichi, NumPy
- Tools and Platforms:** Linux, Git